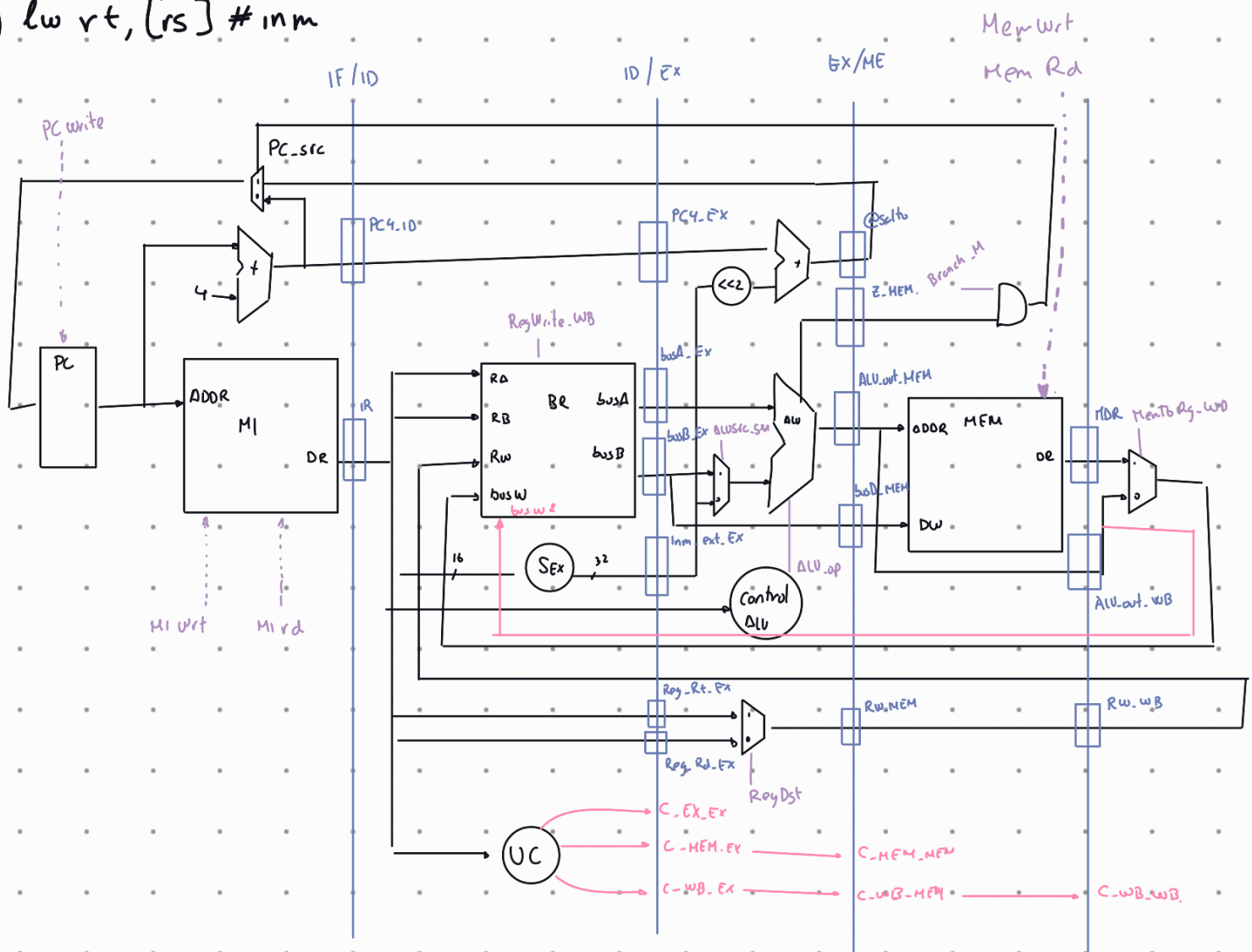


1) lw rt, [rs] # in m



lw rt, [rs] #imm

$$BR(rt) \leftarrow Mem(BR(rs))$$
$$BR(rs) \leftarrow BR(rs) + \text{Sig Ext (lnm 16)}$$
$$P_C \leftarrow P_{C+4}$$

<u>IF</u>	<u>ID</u>	<u>EX</u>	<u>MEM</u>	<u>WB</u>
$PC \leftarrow PC + 4$	$\Delta \leftarrow BR[rs]$ $B \leftarrow BR[rt]$ $INH \leftarrow SigPrt(inn16)$	$ALU_{out} \leftarrow \Delta + Inn$	$LMD \leftarrow Mem[ALU_{out}]$	$BR[rt] \leftarrow LMD$ $BR[rs] \leftarrow ALU_{out}$
	$IR = Mem[PC]$ $PC4_ID : PC + 4$	$PC4_EX : PC4_ID$ $busA_EX : BR[rs]$ $busB_EX : BR[rt]$ $Inn.Prt_EX : SigExt(inn16)$ Reg_Rt_EX Reg_Rd_EX	$@Solto : PC4_EX$ $+inn_ext_EX$ $ALU_{out}.WB = ALU_{out}.MEM$ $Z.MEM : 0$ $ALU_{out}.MEM = busA_EX$ $+inn_ext_EX$ $busB.MEM : busB_EX$ $RW.MEM.Rt : Rd$	$MOR = Mem[ALU_{out}.MEM]$ $RW.WB : RW.MEM$

PC write = 0
 M1 write = 1

Reg Write WB = 0

ALU src sex = 0
 ALU - op = 00 (sum)
 Reg - dest = -

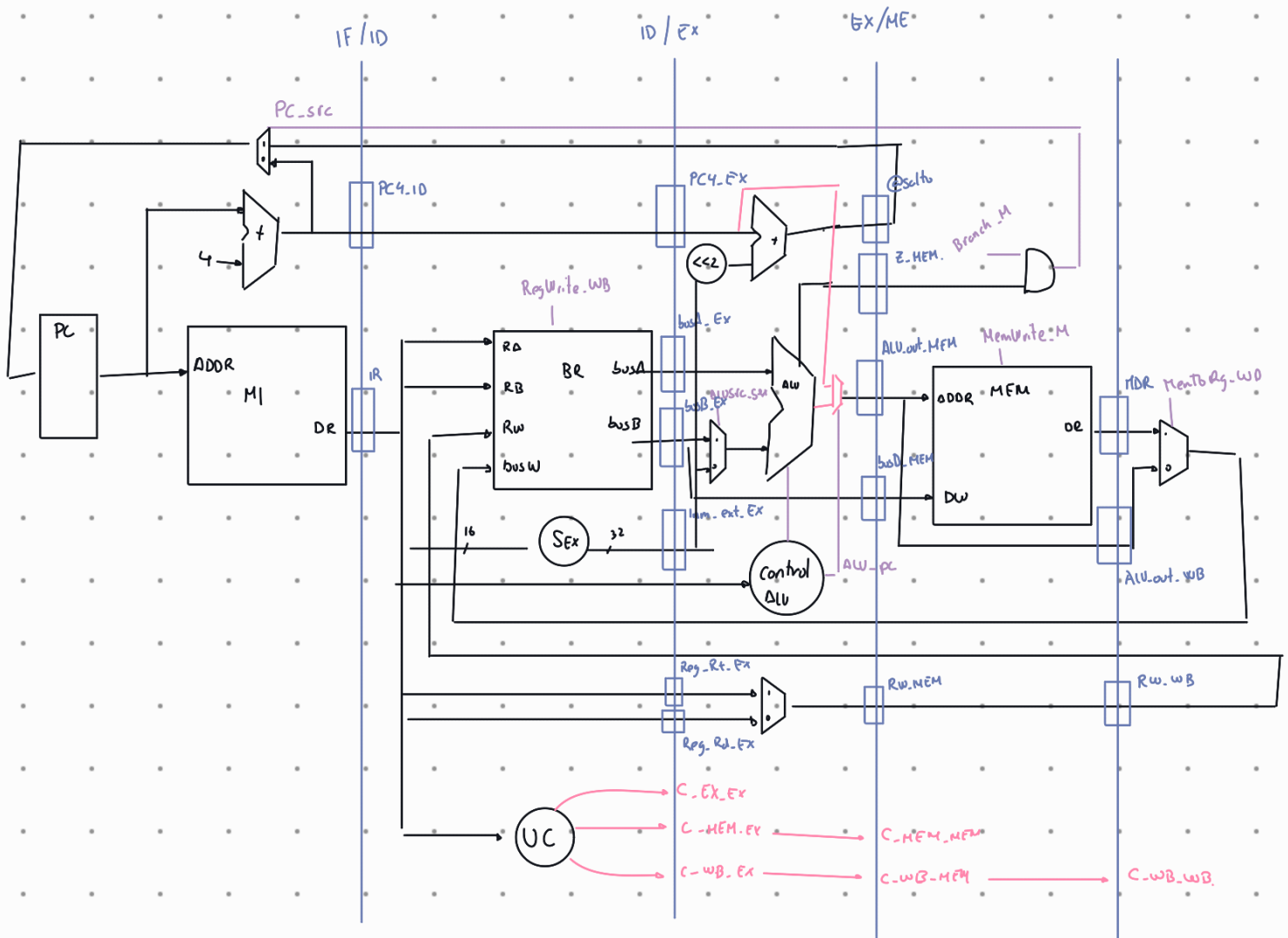
branch_m = 0
 Mem write = 0
 Mem read = 1

Mem to RA = 1

11) jal rt, imm16

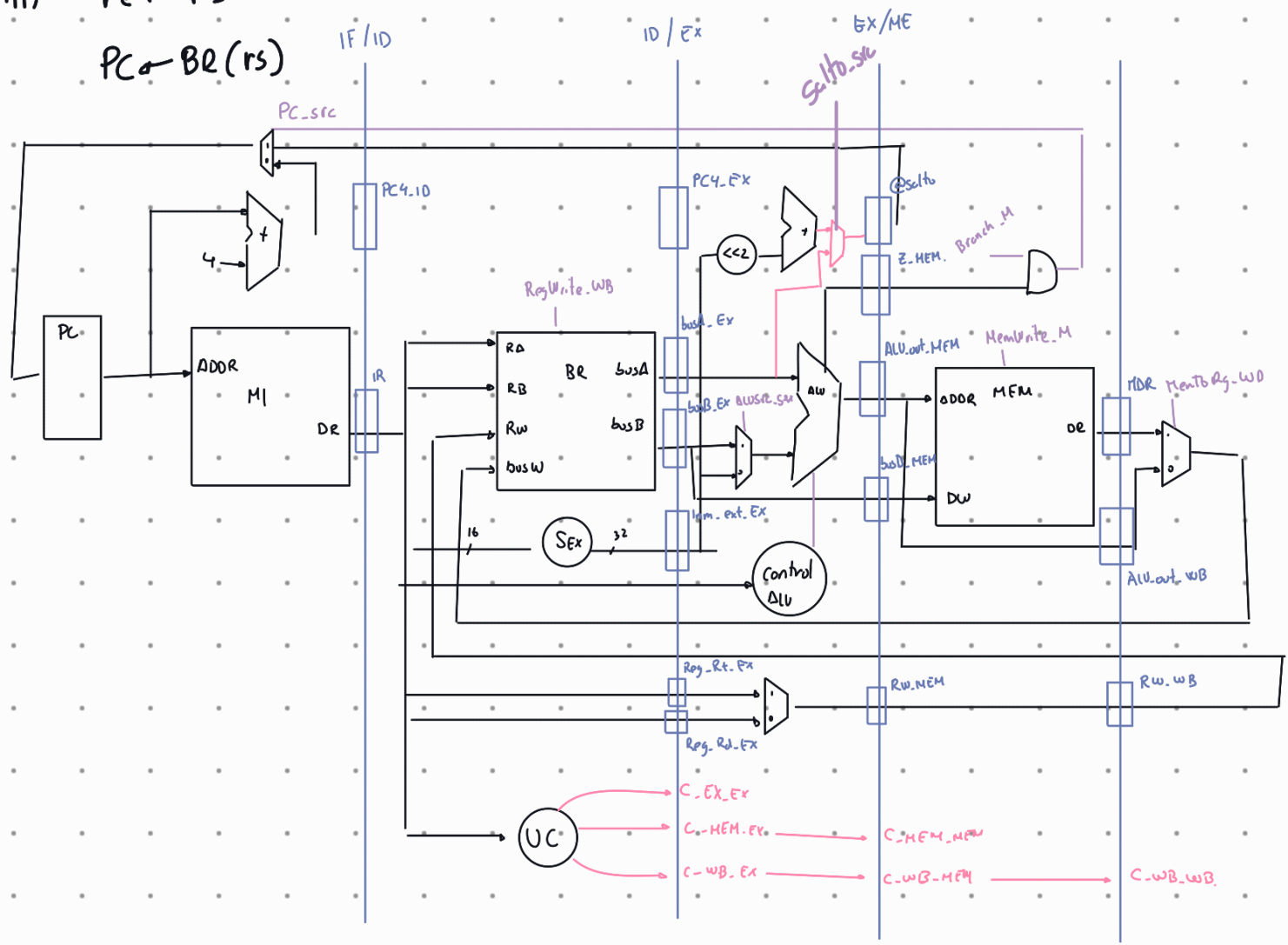
$BR(rt) \leftarrow PC + 4$

$PC \leftarrow PC + 4 + 4 * \text{Sig Ext}(imm16)$



IF	ID	EX	MEM	WB
$IR \leftarrow Mem[PC]$ $PC \leftarrow PC + 4$	$A \leftarrow BR[rs]$ $B \leftarrow BR[rt]$ $Imm \leftarrow SigExt(imm/16)$ $IR = Mem[PC]$ $PC4_ID : PC + 4$	$PC4_EX = PC4_ID$ $busA_EX = BR[rs]$ $busB_EX = BR[rt]$ $Imm_Ext_EX = SigExt(imm/16)$ Reg_Rt_EX Reg_Rd_EX	$PC \leftarrow PC + Imm * 4$ $@Selto : PC4_EX + Imm_Ext_EX$ $Z_Mem = 1$ $ALUout_MEM = PC + 4$ $busB_MEM = busB_EX$ $Rw_MEM : Rt \circ Rd$	$BR[rt] \leftarrow ALUout$ $MOR = Mem[ALUout]$ $ALUout_WB = ALUout_MEM$ $Rw_WB = Rw_MEM$
$PCwrite = 0$ $MIwrite = 1$	$RegWriteWB = 0$	$ALUsrc_EX = -$ $ALU_op = -$ $Reg_dest = -$ $ALU.pc = 1$	$branch_M = 1$ $Memwrite = 0$ $Memread = 1$	$MemtoRA = 0$

iii) ret rs
 $PC \leftarrow BR(rs)$



IF $IR \leftarrow Mem[PC]$ $PC \leftarrow PC + 4$

PC write = 0
 Mem write = 1

ID $A \leftarrow BR[rs]$ $B \leftarrow BR[rt]$ $Imm \leftarrow SigExt(imm16)$ $IR = Mem[PC]$ $PC4_ID : PC + 4$

Reg Write WB = 0

EX $PC4_EX : PC4_ID$ $busA_EX : BR[rs]$ $busB_EX : BR[rt]$ $Imm_Ext_EX : SigExt(imm16)$ $Reg_Rt_EX :$ $Reg_Rd_EX :$

$Op_to_src : 0$
 $ALU_src_sel : 1$
 $ALU_op : -$
 $Reg_dest : 1$
 $ALU_pc : 1$

MEM $PC \leftarrow A$ $@Op_to : busA_EX$ $Z_Mem : 1$ $ALU_out_MEM :$ $busB_Mem : busB_EX$ $RW_Mem : Rt \circ Rd$

branch_M = 1

Mem write = -

Mem read = -

WB

MOR =

 $ALU_out_WB :$ $RW_WB :$

Mem to RA = -